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En un espacio de Hilbert la bola unidad es estrictamente convexa

Proposición 1. *Sea H un espacio de Hilbert, entonces*

$$\overline{B}_1(0) = \{x \in H : \|x\| \leqslant 1\} \text{ es estrictamente convexo.}$$

***Demostración*:** La desigualdad de Cauchy-Schwarz es estricta siempre que los vectores no sean linealmente dependientes. Por tanto, para $x, y \in B_1(0)$, $x \neq y$ y $t \in (0,1)$, tenemos

$$\begin{aligned} \|tx + (1-t)y\|^2 &= \langle tx + (1-t)y, tx + (1-t)y \rangle \\ &= |t|^2 \|x\|^2 + |1-t|^2 \|y\|^2 + 2\,\Re\left(t(1-\overline{t})\langle x, y\rangle\right) \\ &< |t|^2 \|x\|^2 + |1-t|^2 \|y\|^2 + 2\ |t|\,|1-t|\,\|x\|\,\|y\| = \left(t\|x\| + (1-t)\|y\|\right)^2. \end{aligned}$$

Como $\|x\|, \|y\| \leqslant 1$, $t\|x\| + (1-t)\|y\| \leqslant 1$ y, por tanto, $\|tx + (1-t)y\| < 1$. ∎

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